Devasmito Das, Ph.D.

Maître de conférences – Université de Montpellier

Research: **LIRMM – DEXTER Team** | Teaching: **Polytech Montpellier – Béziers Campus**

Control Systems • Nonlinear Dynamics • Robotics • Fractional-Order Systems

Google Scholar | ORCID: 0009-0003-0661-2251 | LinkedIn | GitHub

Profile

Control and robotics researcher working at the interface of **nonlinear dynamics, model predictive control, fractional-order systems, chaos and learning-enhanced control**. Current work at LIRMM focuses primarily on the **modelling, estimation and advanced control of soft/continuum robots and cable-driven parallel robots (CDPRs)**, with emphasis on robustness, constraints, safety and real-time implementation. Broader interests include state estimation, adaptive/robust control, synchronization, networked dynamics and physics-informed learning.

Academic Appointment

Maître de conférences 2026–present
Université de Montpellier France

Research at **LIRMM (DEXTER Team)**; teaching at **Polytech Montpellier, Béziers Campus**. Teaching areas include algorithms and programming (Python), programming for robotics, and student/research projects.

Education

Ph.D. in Automation, Control and Robotics
École Centrale de Nantes / Indian Institute of Technology Madras France / India

Joint doctoral programme; research on **coupled networks of fractional nonlinear oscillators**, with emphasis on fractional dynamics, synchronization, chaos, nonlinear and predictive control.

Master's in Control and Robotics
École Centrale de Nantes France

Specialization in **Advanced Robotics**.

Bachelor of Technology (B.Tech.) in Electrical Engineering
Meghnad Saha Institute of Technology India

Research Themes

- **Soft and continuum robotics:** nonlinear modelling, viscoelastic dynamics, state estimation, NMPC/FMPC, robust and real-time control.
- **Cable-driven parallel robots:** adaptive/robust NMPC, unilateral tension constraints, wrench/workspace feasibility, uncertain payloads and safety limits.
- **Fractional and nonlinear dynamics:** identification, stability, chaos suppression, synchronization and memory-aware modelling.
- **Learning-enhanced control:** PINNs, reinforcement learning and self-supervised/physics-informed methods integrated with model-based control.

Selected Research Project

PHC RILA – French-Bulgarian Research Collaboration, Technical University of Sofia 2024–2025

International collaboration on advanced control of nonlinear dynamical systems with **Prof. Simona Petrakieva** and **Prof. Tsonyo Slavov**. Research visit to Sofia (10–18 June 2025) included laboratory visits, research presentation and technical exchanges with researchers and doctoral students. The collaboration generated joint scientific outputs, including contributions presented/published at **IEEE ICARAI 2025 and ICARAI 2026**.

Selected Publications

1. **D. Das**, I. Taralova, T. Slavov, J.-J. Loiseau, M. Pandey, "Frequency domain identification of a 1-DoF and 3-DoF fractional-order Duffing system using Grünwald–Letnikov characterization," *Fractal and Fractional*, 9(9), 581, 2025. DOI
2. **D. Das**, "A Fractional-Order Hyperchaotic Framework for Dynamic Affine S-Box Parameterization and Bidirectional Diffusion with Neural Nonlinear Mixing," *Nonlinear Science*, 100162, 2026. DOI
3. **D. Das**, "Game-Theoretic Nonlinear Model Predictive Control for Chaos Suppression and Orbital Stabilization in Coupled Fractional-Order Duffing Oscillators," *Nonlinear Science*, 100141, 2026. DOI
4. **D. Das**, I. Taralova, J.-J. Loiseau, "Time-delay feedback control of fractional chaotic Rössler oscillator," *IFAC-PapersOnLine*,

58(5), 90–95, 2024. DOI

Selected Publications (continued)

5. **D. Das**, I. Taralova, J.-J. Loiseau, S. Filipova-Petrakieva, "Fractional model predictive control of fractional chaotic Rössler oscillator," *IFAC-PapersOnLine*, 59(11), 156–161, 2025. DOI
6. **D. Das**, I. Taralova, T. Slavov, J.-J. Loiseau, "PINN-Enhanced Constrained NMPC for Chaos Suppression in a Fractional Forced Duffing Oscillator," *2026 International Conference Automatics, Robotics and Artificial Intelligence (ICARAI)*, pp. 1–6. IEEE. DOI
7. **D. Das**, I. Taralova, J.-J. Loiseau, "Fuzzy Adaptive Sliding Mode Control of Fractional-order Chaotic Lorenz Systems," *2026 12th International Conference on Control, Decision and Information Technologies (CoDIT)*, pp. 257–262. IEEE. DOI
8. **D. Das**, I. Taralova, J.-J. Loiseau, T. Slavov, M. Pandey, "Fractional mixed-integer model predictive control of fractional Rössler oscillator," *13th IFAC International Conference on Fractional Differentiation and its Applications (ICFDA)*, 2025. DOI

Selected accepted/in-press work includes IFAC World Congress 2026, IEEE ICIDeA 2026, IEEE iAIMS 2026, IEEE ICIIS 2026, CHAOS/Springer proceedings, and additional journal manuscripts in nonlinear dynamics and control.

Teaching Experience

Université de Montpellier / Polytech Montpellier — 2026–present
Algorithms & Programming (Python); Programming for Robotics; Student and Research Projects — Béziers

École Centrale de Nantes — 2022–2024
Control & Observation; Professional Engineering Projects — 64 h

Laboratory teaching in control/observation (16 h) plus tutoring and evaluation of industry-linked engineering projects (48 h) with partners including **UCNA, EXPLEO, KASADENN, SAINT-GOBAIN, SPIE INDUSTRIE and VEOLIA**.

Indian Institute of Technology Madras — 2024–2025
Design of Machine Elements; Theory of Vibrations; Rotor Dynamics Laboratory — 96 h

Tutorials, assignment/project evaluation and laboratory teaching across undergraduate (L3) and Master's (M2) levels.

Academic Service & Peer Review

- Reviewer for **IFAC World Congress 2026**, **iAIMS 2026**, **ICIDeA 2026**, **ICAISF 2026**, **ICECER 2026**, **ICECET 2026**, **IWACCE 2025**, and the journal *TELKOMNIKA*.
- Review portfolio spans nonlinear/predictive control, fractional systems, system identification, multi-agent systems, AI/XAI, deep learning, renewable energy and engineering applications.
- Scientific outreach: **DECLICS** mediation session with high-school students at Lycée Nicolas Appert; iterative research communication and discussion of scientific and academic careers.

Selected Achievements & Professional Engagement

- **Qualification, CNU Section 61 (France)**; appointed Maître de conférences at Université de Montpellier.
- Joint doctoral training across **France and India**; international research collaborations extending to Bulgaria and Japan.
- **Embedded World 2026, Nuremberg –** participation by invitation from MathWorks; technical exchanges on embedded control, MATLAB/Simulink, electric motor control, biomedical signals and real-time implementation, including interactions with CEA, AMD, Qualcomm, STMicroelectronics and Analog Devices.
- **Global Industrie Paris 2026 –** technology watch and industrial exchanges in robotics, automation, embedded systems, intelligent manufacturing, digital twins and Industry 4.0.
- **MathWorks / MATLAB EXPO 2025 –** hands-on training in reduced-order modelling and reinforcement learning with MATLAB/Simulink.

Technical Expertise

Control: MPC/NMPC/FMPC, nonlinear control, adaptive/robust control, sliding mode, backstepping, feedback linearization, PID/RL-PID, state estimation.

Dynamics: fractional-order systems, nonlinear oscillators, chaos, synchronization, networked systems, Lorenz/Rössler/Duffing-type models.

Robotics & computation: soft/continuum robotics, CDPRs, MATLAB/Simulink, Python, physics-informed learning, reinforcement learning, real-time and embedded-control considerations.